

RESEARCH REPORT

Prediction of disease progression in Miller Fisher and overlap syndromes

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Abstract Patients with Miller Fisher syndrome (MFS) may have a relatively mild clinical course or progress to Guillain-Barré syndrome (GBS) with limb weakness (MFS-GBS overlap syndrome). Other variants in this spectrum are GBS with ophthalmoparesis and Bickerstaff's Brainstem encephalitis (BBE). To compare the clinical course of MFS and overlap syndromes and to identify predictors of disease progression. In a prospective study of 170 patients with GBS and variant forms, 37 (22%) had a MFS, MFS-GBS overlap syndrome, ophthalmoplegic GBS or BBE. The clinical, serological, and electrophysiological features were compared. Twenty-three patients presented with MFS, of which 10 (43%) developed limb weakness (MFS-GBS overlap syndrome). All these transitions occurred in the first week after onset of symptoms. There were no differences in the clinical, electrophysiological and serological features at entry between MFS and MFS-GBS. Twelve patients had ophthalmoplegic GBS and the disease severity at nadir and outcome was worse than in the patients with a MFS-GBS overlap syndrome. No early predictors for progression from MFS to MFS-GBS overlap syndrome were found. All transitions occurred in the first week. This finding implicates that all patients with MFS need careful monitoring for at least 1 week.

Key words: Miller Fisher syndrome, ophthalmoparesis, progression to overlap syndrome

Introduction

Ophthalmoparesis is a distinctive clinical feature in patients with Miller Fisher syndrome (MFS) and Guillain-Barré syndrome (GBS). Patients with MFS have ophthalmoparesis with additional ataxia and areflexia without limb weakness, and usually have a relatively benign clinical course requiring no specific treatment (Mori *et al.*, 2001; Overell *et al.*, 2007). Some patients presenting with MFS may deteriorate and develop additional limb weakness (MFS-GBS overlap syndrome) or bulbar or respiratory failure (Blau *et al.*, 1980; Funakoshi *et al.*, 2009; Sekiguchi *et al.*, 2016).

Ophthalmoplegia could also occur later on in patients presenting with GBS with limb weakness and sensory deficits ("ophthalmoplegic GBS"). Other patients in this related spectrum of disorders have GBS and may later on develop consciousness disturbances and pyramidal tract signs and are diagnosed as Bickerstaff's brainstem encephalitis (BBE) (Wakerley *et al.*, 2014). The clinical course and outcome of these subtypes may vary considerably.

At present, no early characteristics have been identified that can be used to predict whether a patient initially diagnosed as MFS is at risk for progression to MFS-GBS overlap syndrome (Sekiguchi *et al.*, 2016). We aimed to determine risk factors for progression to MFS-GBS overlap syndrome and compared the clinical course and outcome of these variant forms.

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Materials and Methods

Data were derived from a prospective, multicenter cohort study in 170 Dutch patients with GBS, MFS or variant forms, who were included within 2 weeks after onset of symptoms (GRAPH study) (Ruts et al., 2010). The study was approved by an ethical standards committee on human experimentation, and informed consent was obtained from all patients.

For the current study, we selected patients with ophthalmoparesis and divided them into four groups: (1) patients without limb weakness (MFS), (2) patients who developed bilateral limb weakness during disease course (MFS-GBS overlap syndrome), (3) patients with limb weakness as presenting symptom, who later on developed ophthalmoplegia (ophthalmoplegic GBS), and (4) patients with disturbance of consciousness diagnosed as BBE (Wakerley et al., 2014).

Data were collected from study entry and after 6 months follow-up. Potential risk factors for disease progression in MFS to MFS-GBS overlap syndrome analyzed were gender, age, antecedent event, presenting neurological features at entry (including facial weakness, sensory deficits, pain, MRC sum-score, GBS disability score), results of CSF and nerve conduction studies (NCS), and presence of serum anti-GQ1b antibodies. Information on ataxia was collected retrospectively from patient records and defined as presence of coordination problems, positive test of Romberg or ataxic gait.

Cerebrospinal fluid (CSF) was tested for white blood cell count and protein level. Serum was tested for IgM and IgG antibodies to GQ1b (IgM and IgG) by ELISA (Kuijff et al., 2005).

Statistical analysis

We used SPSS (version 21) for statistical analyses. Categorical data were presented as proportions and differences were analyzed using the Fisher's exact test or Chi-squared test. Continuous data were presented as medians and interquartile ranges (IQR) and compared using the Mann-Whitney *U* test. A two-sided *p*-value of <0.05 was considered to be statistically significant.

Results

In the cohort of 170 patients, 38 (22%) had ophthalmoparesis at entry or during disease. One patient was excluded because of an additional transverse myelitis, resulting in 37 patients included in this study. Twenty-three patients presented with MFS, of whom 10 (43%) later on developed limb weakness (MFS-GBS overlap). Twelve patients (32%) had ophthalmoplegic GBS and 2 patients (5%) BBE (Fig. 1 and Table 1).

Limb weakness occurred in all MFS-GBS patients within 1 week after onset of symptoms (median 1 day, range 0–6 days). No significant differences were found in the clinical features at entry that could be used to predict progression from MFS to MFS-GBS overlap syndrome (Table 1).

At entry, patients with ophthalmoplegic GBS had more severe limb weakness (MRC sum score 39, IQR 27–43) than MFS-GBS overlap (MRC sum score 53, IQR 44–58) (*p* = 0.02) and a worse median GBS disability score (median 4 (IQR 4–5) vs. 3 (IQR 3–4)) (*p* = 0.04) (Table 1). Two MFS patients had bulbar weakness at presentation and another patient with MFS developed bulbar weakness after 1 week. None of them required mechanical ventilation. The two BBE patients had the most severe clinical picture, being completely paralyzed and ventilated. Both developed disturbances in consciousness within 12 days after onset of weakness.

A cyto-albuminologic dissociation in CSF was more often found in ophthalmoplegic GBS patients (67%) than in the other groups (MFS 18% and MFS-GBS overlap 20%) (*p* = 0.02).

NCS revealed that patients with MFS most often had sensory nerve involvement only (44%), whereas ophthalmoplegic GBS and patients with MFS-GBS overlap most often had both sensory and motor involvement (respectively 100% and 70%, MFS 33%, *p* = 0.007). All patients with ophthalmoplegic GBS had evidence of motor nerve involvement on NCS: 33%–67% had reduced distal compound muscle action potentials (dCMAP's) or unresponsive nerves, 22%–50% had prolonged distal motor latencies (DML), 34%–67% had decreased motor conduction velocity (MCV), and 50%–100% had abnormal F-waves. In the MFS-GBS overlap group, motor involvement consisted of abnormal dCMAP amplitudes in 0%–75% of the patients, prolonged DML in 13%–38%, decreased MCV in 0%–38%, and prolonged or absent F-waves in 30%–50%. In contrast, MFS patients showed more often sensory nerve involvement (0%–28% abnormal sensory nerve action potential [SNAP], 43%–75% prolonged distal sensory latency [DSL], and 0%–57% decreased sensory conduction velocity [SCV]) and less frequent motor nerve involvement (0%–13% abnormal dCMAP amplitudes, 0%–33% prolonged DML, 0%–34% decreased MCV, and 0% abnormal F-waves).

Serum anti-GQ1b IgG antibodies were more often present in patients with MFS (85%) or MFS-GBS overlap (63%) than in patients with ophthalmoplegic GBS (8%) (*p* < 0.001). One patient with BBE was positive for IgM to GQ1b, but negative for IgG to GQ1b. The other patient with BBE was negative for both IgM and IgG to GQ1b.

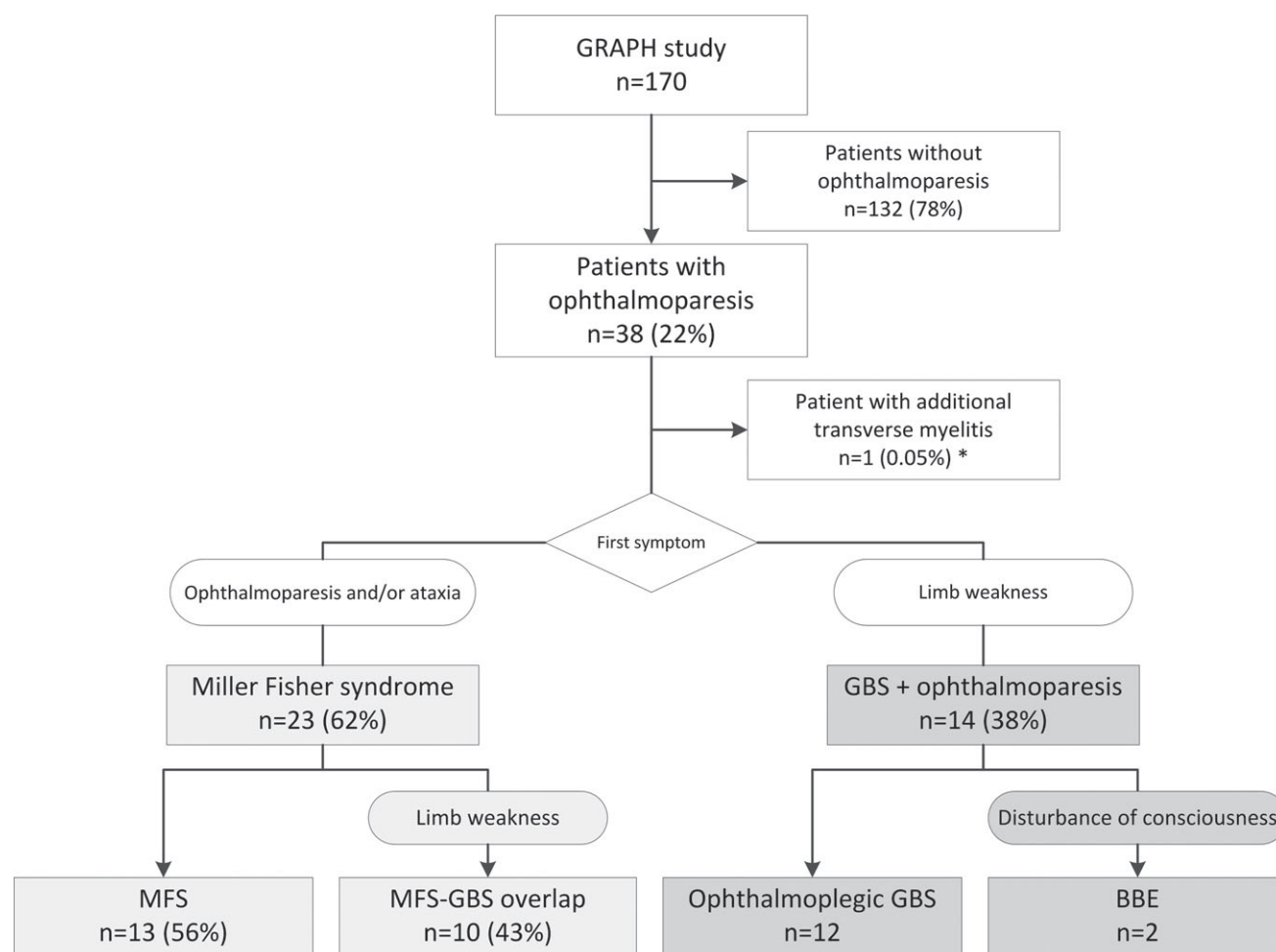


Figure 1. Patient inclusions. BBE, Bickerstaff brainstem encephalitis; GBS, Guillain-Barré syndrome; GRAPH, GBS Research about Pain and Heterogeneity; MFS, Miller Fisher syndrome. *This patient was diagnosed with atypical GBS with transverse myelitis (demyelinating lesion in thoracic spine).

Univariate analysis did not reveal any features at entry which were significantly associated with progression to MFS-GBS overlap (defined as an odds ratio > 1.0 and calculated for sensory deficits, facial weakness, GBS disability score at entry, CSF examination, results of NCS, and presence of GQ1b antibodies).

None of the patients with MFS developed respiratory failure, while 1 patient (10%) of the MFS-GBS overlap group and 7 (58%) of the ophthalmoplegic GBS patients and both BBE patients needed ventilatory support at some time during the disease course (*p*-value MFS-GBS overlap vs. GBS group = 0.03). In the patient with MFS-GBS overlap syndrome, respiratory failure occurred within 1 day after onset of symptoms.

At nadir, the MRC-sum score was lower in the ophthalmoplegic GBS group than in the MFS-GBS overlap group (median 24 [IQR 4–39] vs. 51 [IQR 44–58]) (*p* = 0.002). These patients also had a worse GBS disability score than the MFS-GBS overlap patients (median 5 [IQR 4–5] vs. 3 [IQR 3–4]) (*p* = 0.01).

Five MFS patients (39%) were treated with IVIg of whom two (40%) had residual symptoms or signs after 6 months follow-up. Of the eight untreated MFS patients, five patients had residual symptoms or signs (63%). All patients in the MFS-GBS overlap, ophthalmoplegic GBS and BBE group were treated with IVIg. After 6 months, limb weakness was still present in 10% of the MFS-GBS overlap syndrome patients, 55% of the ophthalmoplegic GBS patients and in both BBE patients. Sensory deficits were found in 38% of the MFS patients, 56% of the patients with MFS-GBS overlap syndrome, 82% of the patients with ophthalmoplegic GBS, and in both patients with BBE patients. Cranial nerve involvement (oculomotor or facial weakness) was observed in 20% of the MFS patients, 44% of the MFS-GBS overlap syndrome patients, 50% of the ophthalmoplegic GBS patients and in both BBE patients.

The median time to independent walking was not significantly different between patients with MFS and

Table 1. Demographical features and clinical course of MFS and overlap syndromes.

	MFS (n = 13)	MFS-GBS overlap (n = 10)	Ophthalmoplegic GBS (n = 12)
Gender (male), n (%)	9/13 (69%)	7/10 (70%)	9/12 (75%)
Age , years (median, IQR)	51 (37–55)	58 (50–64)	47 (39–61)
Antecedent event			
Diarrhea, n (%)	2/13 (15%)	–	3/12 (25%)
Upper respiratory infectious symptoms, n (%)	5/13 (39%)	7/10 (70%)	4/11 (36%)
Clinical features at entry			
Clinical triad*, n (%)	8/13 (62%)	6/9 (67%)	2/10 (20%)
Ophthalmoparesis (at least 1 nerve), n (%)	13/13 (100%)	9/10 (90%)	7/9 (78%)
Ataxia, n (%)	8/13 (62%)	6/8 (75%)	2/10 (20%)
Decreased reflexes or areflexia, n (%)	12/13 (92%)	9/10 (90%)	9/10 (90%)
Paresthesias, n (%)	9/13 (69%)	6/9 (67%)	11/11 (100%)
Pain, n (%)	7/12 (58%)	3/10 (30%)	7/10 (70%)
Sensory deficits, n (%)	5/13 (39%)	7/10 (70%)	10/11 (90%)
Facial weakness, n (%)	5/13 (39%)	5/10 (50%)	6/10 (60%)
MRC-sum score, median (IQR)	60 (60–60)	53 (44–58)	39 (27–43)
GBS-disability score, median (IQR)	3 (2–3)	3 (3–4)	4 (4–5)
Ventilator dependency, n (%)	–	1/10 (10%)	7/12 (58%)
Additional investigations			
Time from onset until LP (days), median, IQR	3 (2–7)	2 (1–4)	4 (1–5)
Cyto-albuminologic dissociation in CSF, n (%)	2/11 (18%)	2/10 (20%)	8/12 (67%)
Time from onset until NCS (days), median (IQR)	5 (3–11)	6 (3–10)	8 (7–24)
Results nerve conduction studies			
Normal, n (%)	2/9 (22%)	–	–
Sensory abnormalities, n (%)	4/9 (44%)	3/10 (30%)	–
Motor (+/– sensory) abnormalities, n (%)	3/9 (33%)	7/10 (70%)	10/10 (100%)
GQ1b-IgG positive in serum, n (%)	11/13 (85%)	5/8 (63%)	1/12 (8%)
GQ1b-IgM positive in serum, n (%)	2/13 (15%)	3/8 (38%)	–

*Clinical triad: involvement of ≥ 1 ocular nerve (oculomotor, trochlear or abducens), ataxia and decreased reflexes or areflexia without limb weakness. LP: lumbar puncture. NCS: nerve conduction study. MFS: Miller Fisher syndrome. MFS-GBS overlap: MFS patients later on developing limb weakness. Ophthalmoplegic GBS: GBS patients (with limb weakness) later on developing ophthalmoparesis.

MFS-GBS overlap (7 vs. 14 days) but was longer in patients with ophthalmoplegic GBS (91 days, p -value MFS-GBS overlap vs. ophthalmoplegic GBS = 0.03). This observation was confirmed in Kaplan-Meijer analysis between the three groups, with a log rank p -value of 0.002. The proportion of patients walking independently at 6 months however was the same for MFS-GBS overlap (80%) and ophthalmoplegic GBS patients (83%), and in addition all MFS patients were able to walk independently.

Discussion

In our study, 43% of the patients with MFS developed limb weakness and one MFS-GBS overlap patient developed respiratory failure. This disease progression occurred in the first week of onset of symptoms in all these cases. We identified no early characteristics to determine which MFS patients are at risk for progression to MFS-GBS overlap syndrome. These findings indicate that at present there are no predictors of disease progression for MFS and that all these patients need careful monitoring of the clinical course for at least 1 week.

Previous studies reported variable proportions of patients with MFS who developed limb weakness ranging from 26% to 50% (Berlit and Rakicky, 1992; Lyu et al., 1997; Mori et al., 2001; Funakoshi et al., 2009; Mori and Kuwabara, 2011; Mori et al., 2012; Sekiguchi et al., 2016). These varying proportions can be explained in part by different definitions of weakness and disease progression. Our study, conducted in the Netherlands, confirmed the findings of a recent study from Japan, showing that about half of the patients with MFS developed limb weakness after the first week of onset of symptoms (Sekiguchi et al., 2016). Respiratory insufficiency in the Japanese MFS-GBS overlap syndrome requiring ventilatory support occurred within the first week after onset of symptoms (Sekiguchi et al., 2016). This is comparable with another study in 45 patients with MFS-GBS overlap syndrome, in whom respiratory insufficiency occurred within 1 week (median 2 days, range 1 to 6 days) (Funakoshi et al., 2009). The current study shows that the same disease dynamics occur in a prospective and unbiased cohort of Dutch patients. Together, these studies demonstrate that patients presenting with MFS have a considerable risk of disease

progression but that this occurs predominantly early in the disease course.

In the current study we defined two subtypes in the MFS-GBS spectrum; the MFS-GBS overlap syndrome and ophthalmoplegic GBS. Both subtypes are characterized by weakness of the limbs and eye muscles, but the sequence in which these symptoms develop differs. Interestingly, our patients with ophthalmoplegic GBS were more severely affected at nadir and had poorer outcome than patients with MFS-GBS overlap syndrome. Apparently, the site weakness appears initially, usually is most severely affected at nadir and shows the slowest and incomplete recovery. This finding also gives further support to the hypothesis that GBS and MFS form a continuous spectrum in which the MFS-GBS overlap syndrome and ophthalmoplegic GBS are intermediate forms (*Wakerley et al., 2014*).

Although MFS is considered to have a benign course, we found that 63% of untreated and 40% of the IVIg treated MFS patients had residual symptoms and signs after 6 months follow-up. No randomized controlled trial has been performed investigating the efficacy of IVIg or plasma exchange in MFS or MFS-GBS overlap syndrome (*Overell et al., 2007*).

The main limitation of the study is the limited number of patients of specific subtypes that has influenced the study power to demonstrate risk factors for disease progression. The Japanese study investigating the difference between the subtypes in 60 patients, also did not identify such risk factors for MFS (*Sekiguchi et al., 2016*).

Disclosure

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C. V. and H. B. did conceptualization of the study, analysis and interpretation of the data, and drafting and revising the manuscript. L. R. has collected data, interpretation of the data and revising the manuscript. P. A. v. D. interpretation of the data and revising the manuscript. B. C. J. has design and conceptualization of the study, analysis and interpretation of the data, drafting and revising the manuscript.

References

- Berlit P, Rakicky J (1992). The Miller Fisher syndrome. Review of the literature. *J Clin Neuroophthalmol* 12:57–63.
- Blau I, Casson I, Lieberman A, Weiss E (1980). The not-so-benign Miller Fisher syndrome: a variant of the Guillain-Barre syndrome. *Arch Neurol* 37:384–385.
- Funakoshi K, Kuwabara S, Odaka M, Hirata K, Yuki N (2009). Clinical predictors of mechanical ventilation in Fisher/Guillain-Barre overlap syndrome. *J Neurol Neurosurg Psychiatry* 80:60–64.
- Kuijff ML, van Doorn PA, Tio-Gillen AP, Geleijns K, Ang CW, Hooijkaas H, Hop WC, Jacobs BC (2005). Diagnostic value of anti-GM1 ganglioside serology and validation of the INCAT-ELISA. *J Neurol Sci* 239:37–44.
- Lyu RK, Tang LM, Cheng SY, Hsu WC, Chen ST (1997). Guillain-Barre syndrome in Taiwan: a clinical study of 167 patients. *J Neurol Neurosurg Psychiatry* 63:494–500.
- Mori M, Kuwabara S (2011). Fisher syndrome. *Curr Treat Options Neurol* 13:71–78.
- Mori M, Kuwabara S, Fukutake T, Yuki N, Hattori T (2001). Clinical features and prognosis of Miller Fisher syndrome. *Neurology* 56:1104–1106.
- Mori M, Kuwabara S, Yuki N (2012). Fisher syndrome: clinical features, immunopathogenesis and management. *Expert Rev Neurother* 12:39–51.
- Overell JR, Hsieh ST, Odaka M, Yuki N, Willison HJ (2007). Treatment for Fisher syndrome, Bickerstaff's brainstem encephalitis and related disorders. *Cochrane Database Syst Rev* 1:CD004761. <https://doi.org/10.1002/14651858.CD004761.pub2>.
- Ruts L, Drenthen J, Jongen JL, Hop WC, Visser GH, Jacobs BC, van Doorn PA, Dutch GBSSG (2010). Pain in Guillain-Barre syndrome: a long-term follow-up study. *Neurology* 75:1439–1447.
- Sekiguchi Y, Mori M, Misawa S, Sawai S, Yuki N, Beppu M, Kuwabara S (2016). How often and when Fisher syndrome is overlapped by Guillain-Barre syndrome or Bickerstaff brainstem encephalitis? *Eur J Neurol* 23:1058–1063.
- Wakerley BR, Uncini A, Yuki N, Group GBSC (2014). Guillain-Barre and Miller Fisher syndromes – new diagnostic classification. *Nat Rev Neurol* 10:537–544.